Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

How many isomers of $C_5H_{11}OH$ will be primary alcohols?

(A) 5

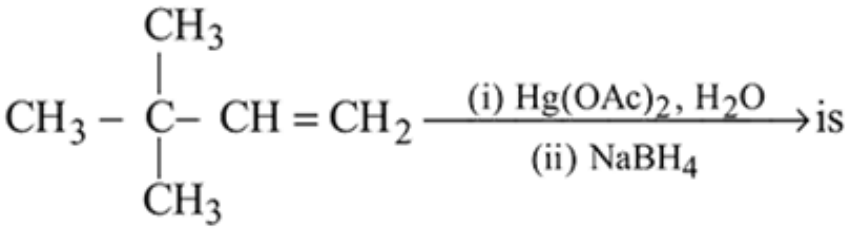
(B) 4

(C) 2

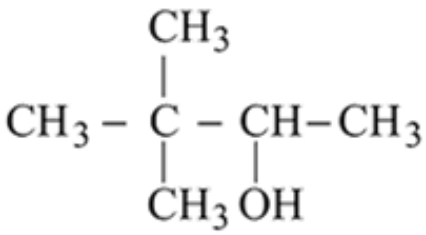
(D) 3

Q.2

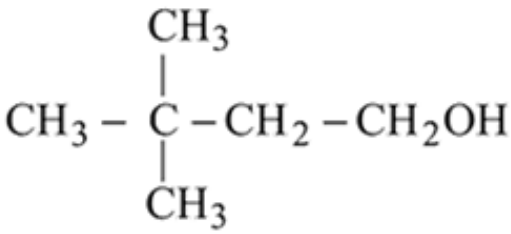
The product of the following reaction,



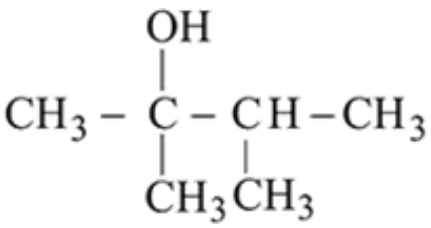
(A)



(B)

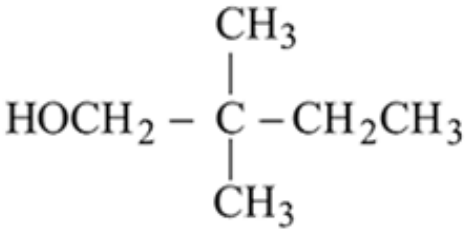


(C)



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(D)



Q.3

Which of the following synthesis gives 3 -methyl-1- hexanol?

(A) 2 - bromohexane $\xrightarrow[C_2H_5OC_2H_5]{Mg}$ $\xrightarrow[(ii)\ H_3O^+]{(i)\ H_2C=O}$

(B) 2 - bromopentane $\xrightarrow[C_2H_5OC_2H_5]{Mg}$ $\xrightarrow[(ii)\quad H_3O^+]{(i)\ \text{  }}$

(C) 3 - bromopentane $\xrightarrow[C_2H_5OC_2H_5]{Mg}$ $\xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3CH=O}$

(D) 1 - bromobutane $\xrightarrow[C_2H_5OC_2H_5]{Mg}$ $\xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3COCH_3}$

Q.4

In the following reaction.

$C_2H_5OC_2H_5 + 4H \xrightarrow{\text{Red P+HI}} 2X + H_2O;$

X is

(A) ethane

(B) ethylene

(C) butane

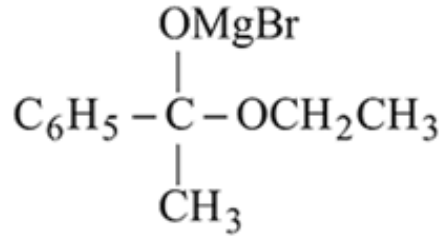
(D) propane

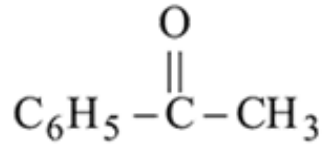
Q.5

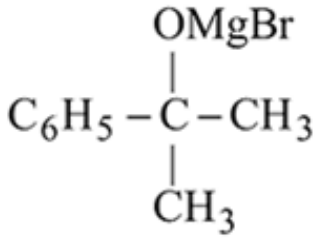
Which of the following are intermediates in the reaction of excess of CH_3MgBr with $C_6H_5COOC_2H_5$ to

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make 2 -phenyl -2 -propanol?

A. 

B. 

C. 

(A) A and B

(B) A, B and C

(C) A and C

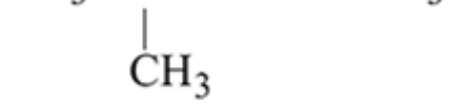
(D) B and C

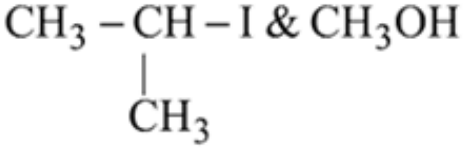
Q.6

$$CH_3 - CH = CH_2 \xrightarrow[\text{(ii) } NaBH_4]{\text{(i) } Hg(OAc)_2/H_2O} X + Na \longrightarrow Y + CH_3Cl \longrightarrow Z + HI \xrightarrow{0^\circ C} A + B$$

What are A and B?

(A) $CH_3 - CH_2 - CH_2 - OH \& CH_3I$

(B) 

(C) 

(D) $CH_3 - CH_2CH_2I \& CH_3OH$

Q.7

Which one/ones of the following reactions will yield 2-propanol?

I. $CH_2 = CH - CH_3 + H_2O \xrightarrow{H^+}$

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II.

$$CH_3-CHO \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgI}$$

III.

$$CH_2O \xrightarrow[(ii)\ H_2O]{(i)\ C_2H_5MgI}$$

IV. $CH_2=CH-CH_3 \xrightarrow{\text{Neutral } KMnO_4}$

(A) I and II

(B) II and III

(C) III and I

(D) II and IV

Q.8

Which of the following has lowest boiling point?

(A) p -Nitrophenol

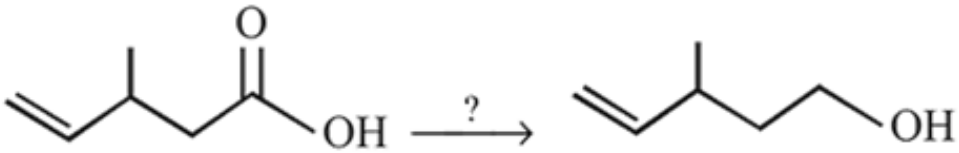
(B) m -Nitrophenol

(C) O -Nitrophenol

(D) Phenol

Q.9

The following change can be carried out with



(A) $NaBH_4$

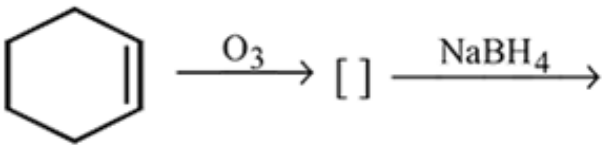
(B) $LiAlH_4$

(C) H_2/Pt

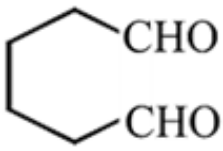
(D) PCC/CH_2Cl_2

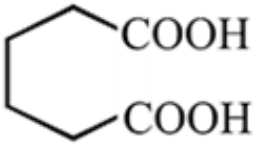
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Q.10

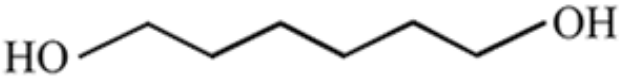


Product is

(A) 

(B) 

(C) mixture of (a) and (b)

(D) 

Q.11

Ethanol can be prepared more easily by which reaction?

(i) $CH_3CH_2Br + H_2O \longrightarrow CH_3CH_2OH$

(ii) $CH_3CH_2Br + Ag_2O$ (in boiling water) $\longrightarrow CH_3CH_2OH$

(A) by (i) reaction

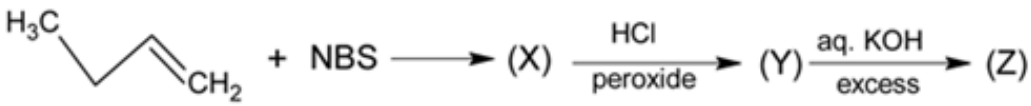
(B) by (ii) reaction

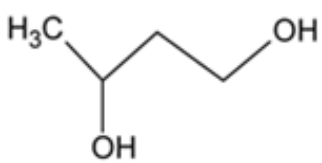
(C) Both reactions proceed at same rate

(D) by none

Q.12

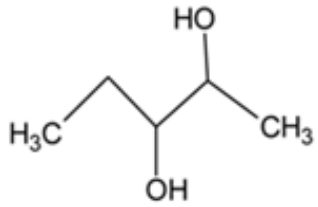
Identify the product [Z] in the given sequence of reaction



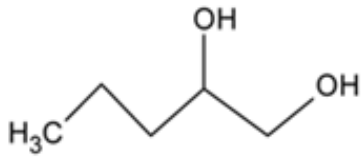
(A) 

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(B)

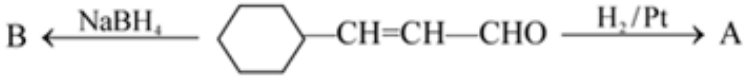


(C)

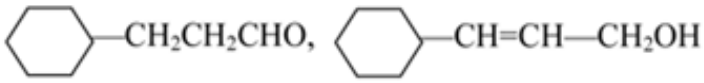


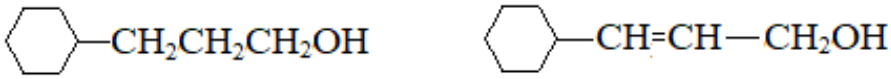
(D) None of these

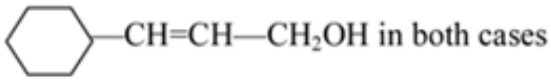
Q.13

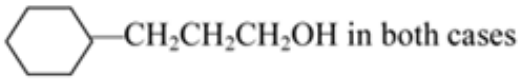


A and B are:

(A) 

(B) 

(C) 

(D) 

Q.14

Dipole moment of
$CH_3CH_2CH_3(I), CH_3CH_2OH(II)$ and $CH_3CH_2\ F(III)$ is in order

(A) I < II < III

(B) I>II>III

(C) I < III < II

(D) III < I < II

Q.15

The boiling points of isomeric alcohols follow the order

(A) primary > secondary > tertiary

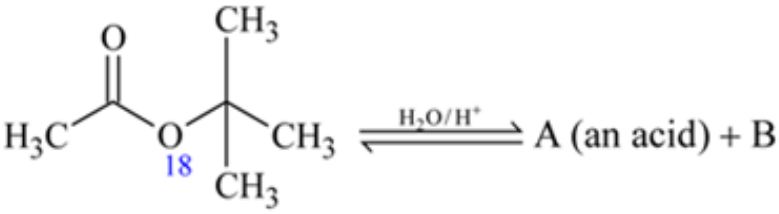
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(B) tertiary > secondary > primary

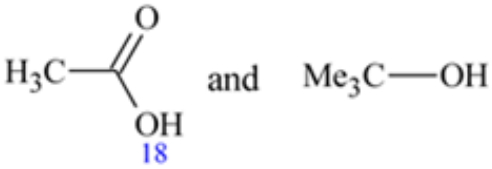
(C) secondary > tertiary > primary

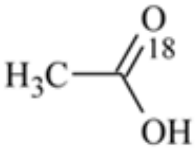
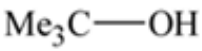
(D) does not follow any order

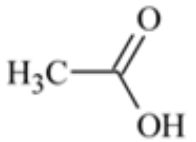
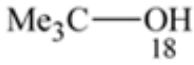
Q.16

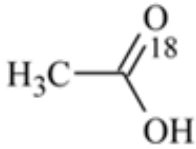
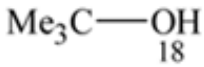


Where A and B are, respectively,

(A)  and

(B)  and 

(C)  and 

(D)  and 

Q.17

Which of the following are isomers

(A) Methyl alcohol and dimethyl ether

(B) Ethyl alcohol and dimethyl ethe

(C) Acetone and acetaldehyde

(D) Propionic acid and propanone

Q.18

Absolute alcohol (100\% alcohol) is prepared by distilling rectified spirit over

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(A) Na

(B) $CaCl_2$

(C) Mg

(D) $Mg(OC_2H_5)_2$

Q.19

Denaturation of alcohol is the

(A) mixing of $CuSO_4$ (a foul smelling solid) and pyridine (to give the colour) to make the commercial alcohol unfit for drinking.

(B) mixing of $CuSO_4$ (to give the colour) and pyridine (a foul smelling solid) to make the commercial alcohol unfit for drinking.

(C) mixing of $Cu(OAc)_2$ and ammonia to make the commercial alcohol unfit for drinking.

(D) mixing of $Cu(OAc)_2$ and pyridine to make the commercial alcohol unfit for drinking.

Q.20

Widespread deaths due to liquor poisoning occurs due to

(A) presence of carbonic acid in liquor

(B) presence of ethyl alcohol in liquor

(C) presence of methyl alcohol in liquor

(D) presence of lead compounds in liquor

Answers & Solutions

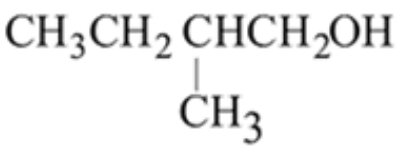
Q.1 Answer:

4

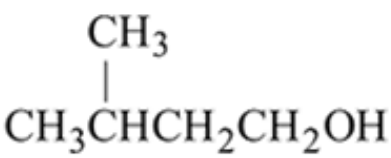
Solution:

(i) $CH_3CH_2CH_2CH_2CH_2OH$

(ii)



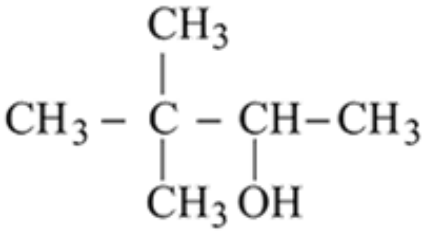
(iii)



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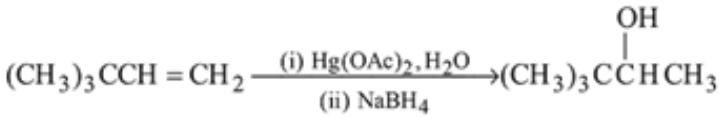
(iv) $(CH_3)_3CCH_2OH$

Q.2 Answer:

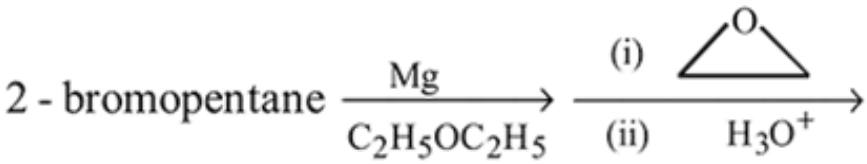


Solution:

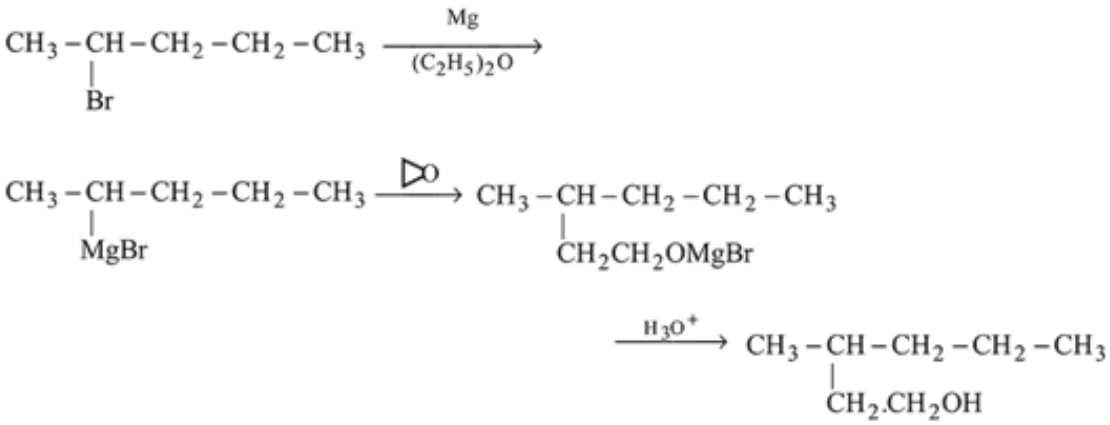
Recall that oxymercuration-reduction of alkenes lead to hydration in Markovnikov's way. Thus



Q.3 Answer:



Solution:



Q.4 Answer:

ethane

Solution:

Ethers are reduced by red P and HI to alkanes through alkyl iodides

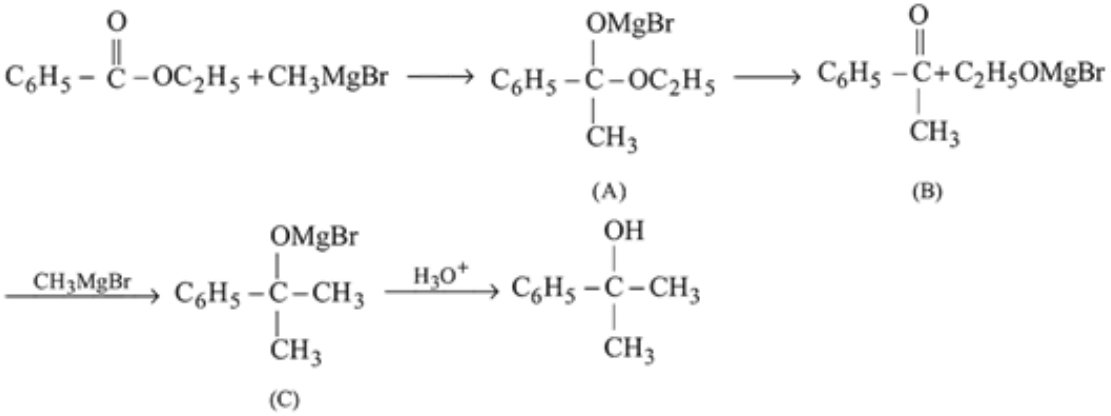
$$C_2H_5OC_2H_5 \xrightarrow{\text{red P+HI}} 2C_2H_5I \xrightarrow{\text{red P+HI}} 2C_2H_6$$

Q.5 Answer:

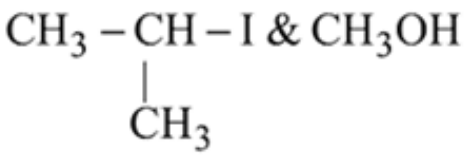
A, B and C

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Solution:



Q.6 Answer:



Solution:

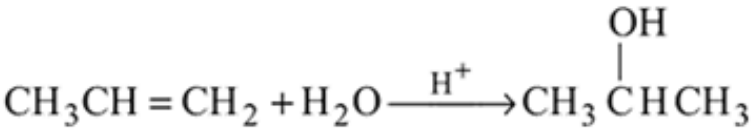
$$CH_3CH=CH_2 \rightarrow CH_3CH(OH)CH_3 \xrightarrow{Na} CH_3CHONaCH_3 \xrightarrow{CH_3Cl}$$
$$CH_3CH(OCH_3)CH_3 \xrightarrow[HI]{^\circ C} CH_3C(CH_3)HI + CH_3OH$$

Q.7 Answer:

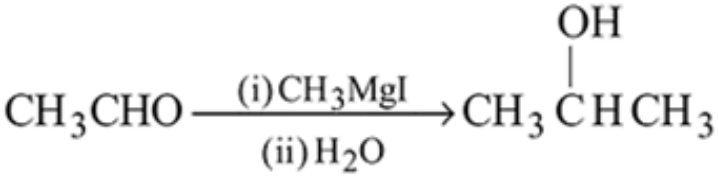
I and II

Solution:

I.



II.



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III.

$$CH_2O \xrightarrow[(ii)\,H_2O]{(i)\,C_2H_5MgI} C_2H_5CH_2OH$$

IV.

$$CH_2CH=CH_2 \xrightarrow[\text{neutral}]{KMnO_4} CH_3\underset{OH}{C}H\underset{OH}{C}H_2$$

Q.8 Answer:

O -Nitrophenol

Solution:

***o*-Nitrophenol has intramolecular H-bonding.**

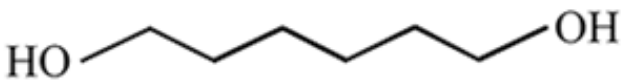
Q.9 Answer:

$LiAlH_4$

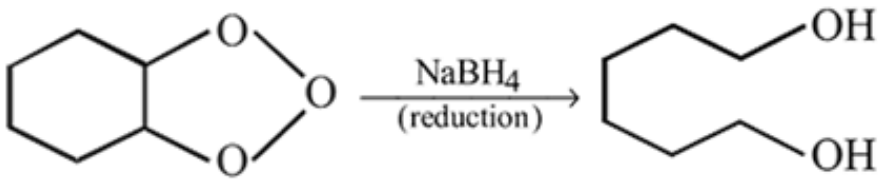
Solution:

Carboxylic acids can be converted to 1° alcohols using Lithium aluminum hydride $(LiAlH_4)$. An aldehyde is produced as an intermediate during this reaction, but it cannot be isolated because it is more reactive than the original carboxylic acid.

Q.10 Answer:



Solution:



Ozonide

Q.11 Answer:

by (ii) reaction

Solution:

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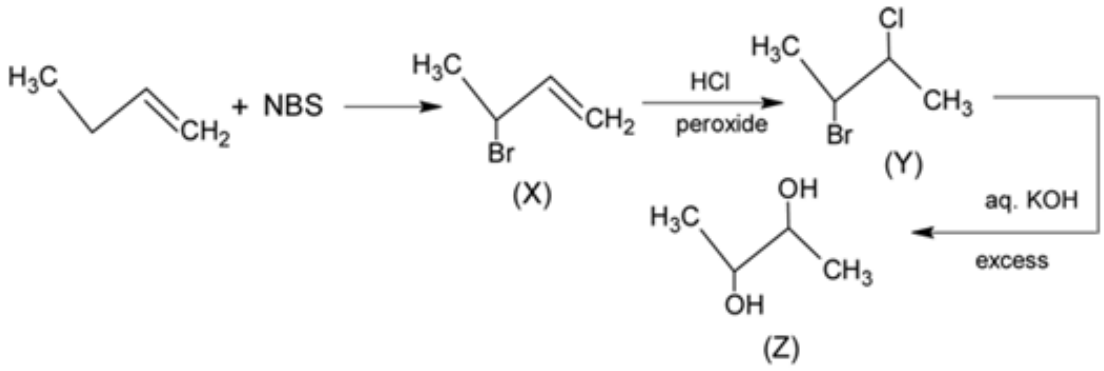
Heavy metal ions, particularly Ag^+, catalyse S_N1 reaction because of the presence of empty orbital.

$$CH_3CH_2Br + Ag^+ \rightleftharpoons [CH_3CH_2 \cdots X \cdots Ag]^+ \xrightarrow[(-AgX)]{\text{slow}} CH_3CH_2^+ \xrightarrow[\text{from } H_2O]{OH^-} CH_3CH_2OH$$

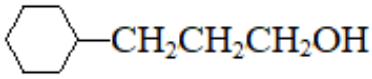
Q.12 Answer:

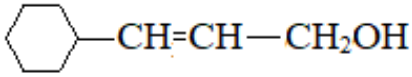
None of these

Solution:



Q.13 Answer:





Solution:

H_2/Pt reduces the doulbe bond also

$NaBH_4$ clearly reduces carbonyl group, protecting the double bond.

Q.14 Answer:

I < II < III

Solution:

Higher the electronegativity between C-X, higher will be the dipole moment.

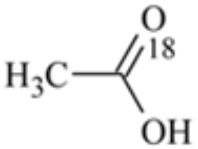
Q.15 Answer:

primary > secondary > tertiary

Solution:

Vander Waal forces are responsible for boiling point.

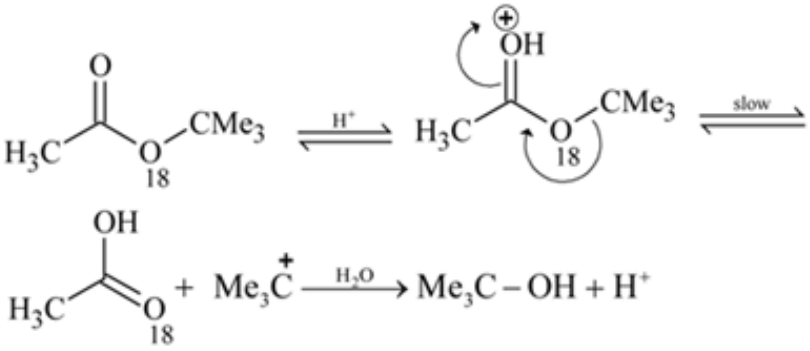
Q.16 Answer:



and $Me_3C—OH$

Solution:

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The above mechanism of ester hydrolysis is known as alkyl-oxygen cleavage (unimolecular).

Q.17 Answer:

Ethyl alcohol and dimethyl ethe

Solution:

CH_3CH_2OH **and** CH_3OCH_3

Q.18 Answer:

$Mg\,(OC_2H_5)_2$

Solution:

$$(C_2H_5O)_2Mg + H_2O \xrightarrow{\text{distill}} \underset{\text{(distillate)}}{2C_2H_5OH} + \underset{\text{(residue)}}{Mg(OH)_2}$$

Q.19 Answer:

mixing of $CuSO_4$ (to give the colour) and pyridine (a foul smelling solid) to make the commercial alcohol unfit for drinking.

Solution:

The commercial alcohol is made unfit for drinking by mixing in it some copper sulphate (to give it colour) and pyridine (a foul smelling liquid). It is known as denaturation of alcohol.

Q.20 Answer:

presence of methyl alcohol in liquor

Solution:

Due to presence of methyl alcohol in liquor.

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